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Project Mímir: Autonomous Rover Swarm with Image-Based Targeting

(Aug 2026 - Present)

The Goal:

Build a set of autonomous rovers that can search for and track a target by an identifiable feature. In doing so also mapping out the space, communicating between one another, and, if needed, operating in a decentralized fashion. I want to keep each rover under 30 dollars and quick to assemble or modify.

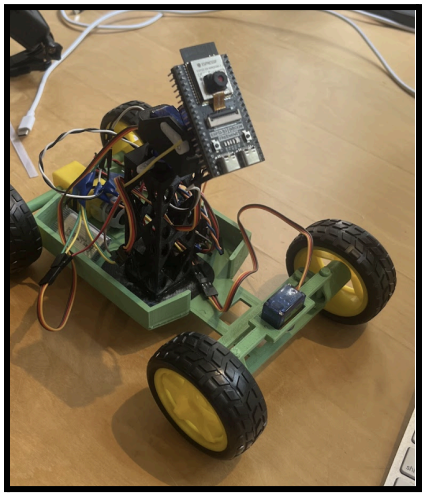
The Execution:

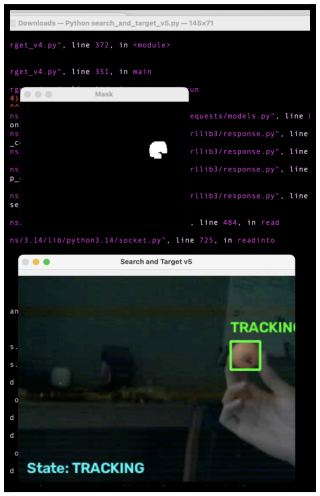
Likely the most important decision I made was selecting the right microcontroller. I chose to use the ESP32-S3 CAM for one main reason: it has the camera, wifi, and compute all in one board while also having enough RAM to handle the QVGA streaming. It also has ESP-NOW, which is a wireless communication protocol that allows the rovers to send packets amongst each other via 2.4 GHz—an elegant solution for decentralized operation. Oh, and it's small and cheap.

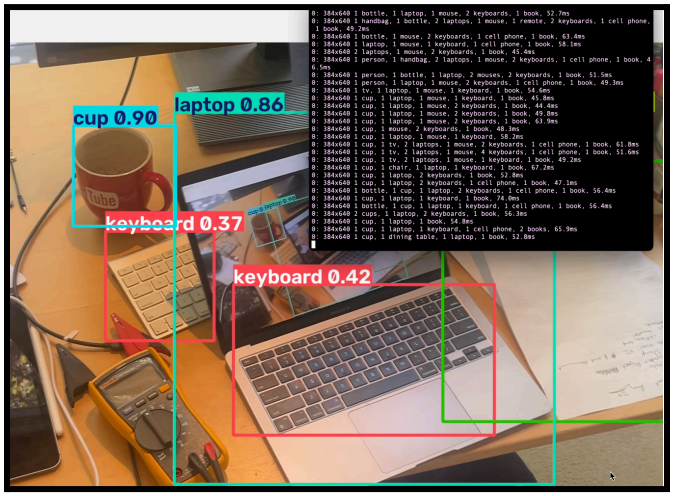
For the first prototype I built the system on top of an ultra-simple 3D printed car—one motor, one servo. I designed a central tower to house the main assembly. One of my prouder moments of the project: I designed it to be modular from the very start... It came in handy later. Atop the tower is the pan and swivel assembly (also all 3D printed). Now the rover could look and drive around. I quickly realized my naivete: mapping a space in 3D using 320x240 in a useful manner is nearly impossible. After a few minutes of thinking, I pivoted to an ultrasonic sensor that I had lying around. I designed a panning servo mount, exploiting my modular tower design to position the system as to not impede the main camera's hemispherical view.

I have been experimenting with YOLOv8n as the centralized vision model. It is small enough to run locally on my M1 Macbook Air. Very impressively it does an excellent job (see third image). I can choose to search for a particular object or even a person, all in around 50ms per frame. It's impressive how far computer vision has come... That is certainly not to say that it's infallible: mistaking my multimeter as a phone or a red poster as a stop sign. It reminds me of AI, Autonomy, and The Future of Warfare: a class I took at Stanford. To what degree can we rely on vision algorithms?

The most educational part was refining the HSV/Contour algorithm. It took me days of testing and debugging. Being able to identify targets in an image with so little compute is really cool. It's the largest image detection program that can run locally on this type of ESP32, thus the only way the decentralized mode can function. The HSV detection algorithm is helpful in communicating the details of a target, once located; those few values are informationally dense and can be easily shared with ESP-NOW.







Next Developments:

- Develop additional capability. For example, I am working on an ESP32-based mini-drone that uses its own PCB as the frame. There are similar concepts online. Here, it would integrate seamlessly because of the ESP32.
- The stream lags at times while in motion due to the integrated PCB antenna: Moving to an external one, and possibly shifting to an ESP32-S3 EYE to run facial recognition model locally.
- Build a targeting memory: a mode where all drones save every location and an image each time target candidates are found (not just when they are investigated), saving them for later human (or higher compute) audit.

Project Bluesmobile: Ultra-Capable 3D Printed RC Car

(2021 - 2026)

The Goal:

Build a nimble, fast, durable, and off-road capable 1:10 scale RC car... Fully 3D printed, of course. Later goals included a top speed record and making it autonomous (i.e. dummy proof).

The Execution:

This is one of my first projects that involved 3D printing. I was 13 years old. I have worked on it, however, for the last 6 years. In that time, I have dispensed immeasurable effort and resources into optimizing this RC car's capabilities. I will never be able to recount all that I have designed, tested, or macgyvered. Here's the TLDR.

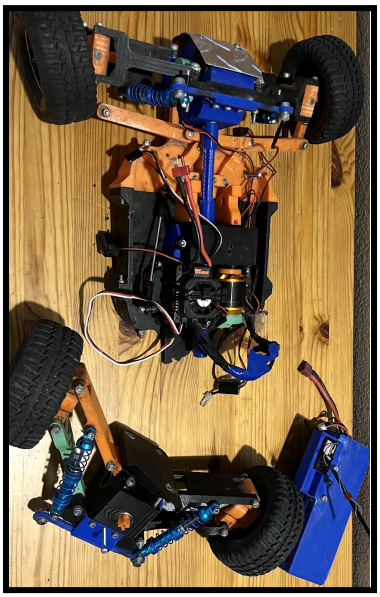
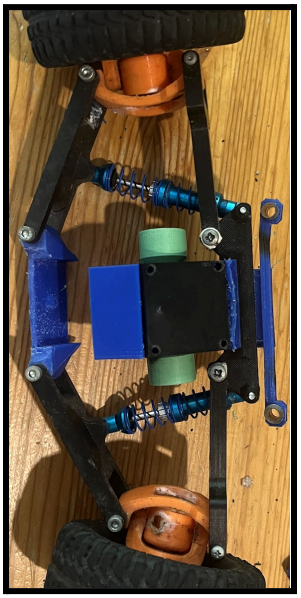
Initially, I pulled publicly available designs from the internet, cobbling them together. I quickly assembled it and took it for a drive. That didn't last long. Not only were there ample mechanical issues, the 3D printed parts were practically disintegrating. Like any educational project, it was back in my room getting disassembled in just a few minutes.

Eventually, I solved the strength issues, but the car still suffered greatly when I inevitably crashed. I used an Arduino Uno and 5 HC-SR04 ultrasonic sensors—designing a bumper to house it all—to make it autonomous. Building a coordinate plane, it could drive toward a location without running into walls and most objects. A clever idea I built in was an ultrasonic sensor 90-degrees from the direction of motion so the car could simply drive along a wall. As one can imagine, this got boring quickly. I changed my code to override human input instead, stopping me from being reckless.

Like many kids, I uncovered a need for speed. Redesigning the double-helical gears and gearbox, the gear ratio went from 1:2 to 1:1—a serious alteration for a car already capable of 25 mph. I had no way to measure speed so I did the obvious... Encased my phone in styrofoam and zip-tied it down tightly :-) . To my satisfaction, it clocked 46 mph that day.

What does the car look like today? The 1000KV D3542 motor drives a 1:2 double-helical central gearbox which spins the two TPU drive shafts. The TPU's flex cleverly prevents the 3D printed gears from stripping. A locked rear differential and open front send power to the wheels via the four CV axles (due to the double wish-bone suspension). A standard 2.4 GHz FHSS signal transmits user input and all electronic components onboard use pulse width modulation.

I recently spent a large amount of time bulletproofing the car, optimizing it one final time. Before, it sat with a shattered chassis (see second image). I redesigned the steering linkage to improve the turning circle and make room for a new driveshaft. I altered the suspension assembly, front and rear, to put less pressure on the dogbones, decreasing rolling resistance, and improve front traction while accelerating hard. Oh, and I set up some cool lights and a GoPro mount.

Next Developments:

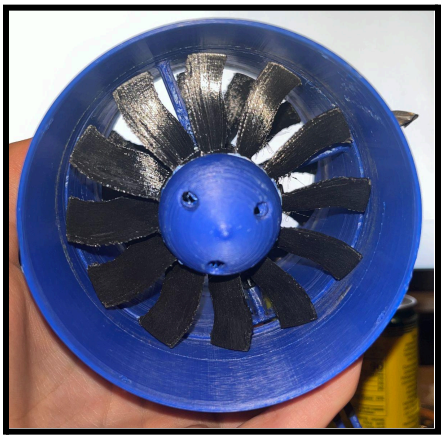
- This is hard to answer for this project. I really feel I've reached a terminus. Taught me a ton, sharpened skills, and, in doing so, kept me entertained for years. As described above, I spent a week of Summer 2026 to return to, repair, and refine the car. It has achieved permanent status in my collection; I won't be scrapping it for parts... Not again, at least.
- That being said, I must credit this car for being an excellent test-bed for new projects. That will not change.

Project Plettenberg: Dual-Prop, Single-Motor Electric Ducted Fan

(Mar 2025 & Sep 2026)

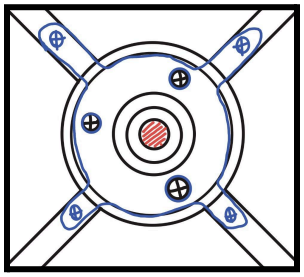
The Goal:

Prototype an ultra-light EDF for my fixed-wing drone project, capable of making many multiples of its own weight in thrust. I want to power both propellers with a single motor.



The Execution:

How does one go about powering 2 propellers with a drone motor (D2826) designed to only spin one? Well I was a bit stumped too. I looked into extended drive shafts that would stick out of both ends. They weren't readily available and with no access to a lathe, it was back to the drawing board. Eventually I decided to try something. Consider the sketch I included below; it shows the butt-end of the motor and how it is attached to the EDF's frame. My solution: disassemble the motor, shift the driveshaft (red) such that it protrudes out of the rear, not the front of the motor. I attached P1 to the front via face screws, and attached P2 via the drive shaft. It left very little room for assembly, but worked as intended.



Another interesting problem: spool 3D printing is far from an ideal process; propellers end up lopsided. As the motor speeds up it would shake the EDF apart. This is a seriously tough engineering problem to solve. I decided on what I deemed an inelegant solution: design the propeller mounts to have "play." Initially it will vibrate, but after a certain speed, the dynamics change; it finds a natural center, aligning its rotation with its principal axis of maximum moment of inertia. I later learned that compressor blades in turbine engines do the same: a tree shaped base allows them to self balance once enough centripetal acceleration is generated... Super cool.

I started it up and began to increase the throttle. I was thrilled by how much power it could produce. Then a little pop, followed by a line of smoke rising into the air and a troubling smell, had me quickly disconnecting the battery. I blew the ESC... a 40 AMP ESC! After some internet sleuthing, I purchased an 80 AMP replacement. Project Plettenberg makes 3x its weight in thrust with printed propellers, which decrease efficiency by 15% to 55% vs injection molded ones according to simulations.

Next Developments:

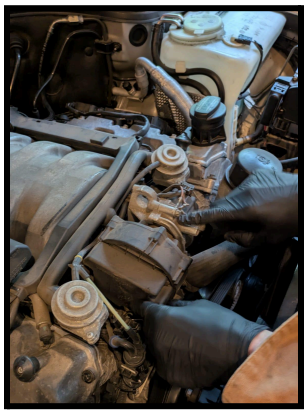
- Integrating this into the fuselage of the fixed-wing drone that I'm making.
- Adding Stationary blades between the props to stop the air from spinning: CFD simulation shows improved thrust output. Fluid dynamics!

Car Repair

(2022 - Present)

The Goal:

In exchange for the freedom to drive my parents' cars, I have (gladly) spent aperiodic weekends working to maintain them: brake pads and rotor replacement, faulty thermostat diagnosis and repair, headlight bulbs, and simple body-work. My biggest success: the hydraulic convertible top mechanism stopped working, the shop wanted $3000 to replace the entire system... I identified a single faulty relay buried deep in the car's belly. It was a $36 job.



I love building stuff. Here are projects that I wish I had more space to expound on. (In no particular order)

- Ultralight Cantilever Truss System: Designed with friends for a competition—awarded second place
- MathWorks Modelling Challenge: Super fun 14-hour sprint to write a 30-page math paper with friends
- Ball Trajectory Prediction Script in Python: HSV tracking, AprilTags, and CoR for bounces (class project)
- 3D Printed Stencil: Ultra-detailed Ent with my brother to promote old-growth conservation.
- 165 psi Compressed Air Launcher: IB Physics capstone project
- Home-Made Driving Simulator: Using potentiometers, limit switches etc. to play Forza
- Arduino Robotic Arm: Basic servo control and PWM
- Using resistance to differentiate buttons: built an electric Piano with a single pin
- Pneumatic Grabber for Robotics Challenge
- Remote Controlled Hotwheels Car: Not an easy design problem
- FRC robotics prototyping: oversized-tennis ball intake, processor, and flywheel launcher
- Rocket Motor: Solid state, potassium nitrate and sugar
- GoPro Screen Replacement: Learned a lot about electronics waterproofing…
- Carbon Fiber Wet-Layup Practice at Past Employer
- Underdesk PC: Compact form-factor for an old computer I built in middle school
- Electronic Shutter Delay for Disposable Camera: 2 hours to prototype, model in Fusion, and assemble
- Spreadsheet to Calculate Spacecraft Temperature from Solar Absorbance and Emittance
- Battery Conversion: Altered the string lights' circuit to support a LiPo… Dorm room upgrade.
- Balsa Wood Glider: With team for class competition, experimented with CFD for design optimization
- Model Suspension Bridge: At 1.5 meters long, made of dowels and paracord, supports 20x its own weight
- … And so many more I simply can't remember.

Research Papers (Class Related - Not Published):

- Technology in Society: Identifying **Domestic Applications of Defense Technology** Currently Being Developed in Ukraine (Written for PWR1MO @ Stanford).
- Math: Using the **conjugate gradient method for non-destructive testing of carbon fiber parts**. Proposed an idea similar to how geophysicists use seismic imaging to find oil reserves.
- Math: Investigating the relationship between personal income and COVID 19 case-rate, death-rate, and mortality. Analyzed 2 years of data from all 3,143 US counties. Identified a moderate, negative relationship between personal income and COVID 19 mortality, which I confirmed to be statistically significant via T-test. It required **large data manipulation**, data cleaning, V-Lookups, Graphing, creating unique identifiers, etc.
- Physics: Formal paper presenting the relationship of increasing wire length and resistance. Built a **clever experimental setup** that allowed for 15 meters of bare copper wire. Was met with a non-standard theoretical equation due to artificial resistance of experimental setup: Propagated uncertainty using partial derivatives.
- Biology: How do differing concentrations of ethanol affect catalase enzymes (commonly found in the liver) with a one-minute incubation period? Measured enzyme activity after incubation by capturing CO_2 released when met with H_2O_2. Identified a linear relationship, however, identified an unexpected, mathematically significant plateau in higher concentrations.